Sandunika Ranasinghe

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RESEARCH INTERESTS

I am interested in leveraging **deep learning, image and signal processing,** and **applied mathematics** to develop theoretically grounded models with strong real-world applicability.

EDUCATION

BSc. (Hons.) in Electrical and Electronic Engineering 2021 March- 2025 August
(First Class Honours) | Academic Record **GPA : 3.95/4.00**
University of Peradeniya, Sri Lanka **Class Rank : 4/115**

G.C.E. Advance level examination 2019
Physics (A), Chemistry (A), Combined Mathematics (A) | Result Sheet
Top 0.8% in Sri Lanka out of 19000+ candidates

PUBLICATIONS

J – JOURNALS, C – CONFERENCES

J 1 **Reflectance Multispectral Imaging for Soil Grain-Size Composition Estimation and USDA Texture Classification**
in *IEEE Access (IF - 4.2)* DOI
S. Ranasinghe[*], S. Jayakody[*], M. De Silva, G. Thilakarathne, R. Godaliyadda, V. Herath, M. P. Ekanayake,S. Navaratnarajah.

C 1 **Trace Energy Regularization for Fidelity Oriented Diffusion Models**
in *15th International Conference on Learning Representations (ICLR 2027)* [Under Review]
S. Ranasinghe, S. Perera, D. Fernando, H.L.P. Malshan, H.P.S. Madushan, R. Godaliyadda, M. P. Ekanayake.

C 2 **Colour-INR - Implicit Neural Representation for Image Colorization through Colour Guidance**
in *15th International Conference on Learning Representations (ICLR 2027)* [Under Review]
B. Apakanda, **S. Ranasinghe**, S. Jayakody, H.D.M.D. Appuhami, R. Godaliyadda, M. P. Ekanayake, R. Ragel.

C 3 **Feature Conditioned Diffusion for Audio Generation**
in *19th International Conference on Industrial and Information Systems 2025 (ICIIS 2025)* DOI
S. Perera, **S. Ranasinghe**, S. Jayakody, B. Epakanda, R. Godaliyadda, M. P. Ekanayake.

C 4 **Deep Learning for Soil Moisture Content Estimation via Reflectance Multispectral Imaging**
in *9th International Conference on Advancements in Technology and Computing 2024 (ICATC 2024)* DOI
S. Ranasinghe, S. Jayakody, M. De Silva, V. Herath, R. Godaliyadda, P. Ekanayake, S. Navaratnarajah, F. Kizel, G. Thilakarathne.

HIGHLIGHTED ACHIEVEMENTS

- Awarded **second runners up** in the **Best Undergraduate Project in Sri Lanka - Manamperi Award 2025** for the Final Year Project on "Multispectral Imaging"
- Awarded the **Silver Award** in the Engineering field at **Sahasak Nimawum Inventions and Innovations Competition 2024** for the Final Year Project on "Multispectral Imaging"
- **Second runners up** in the Healthcare category at **ACES Hackathon 2023**, organized by the Association of Computer Engineering Students, University of Peradeniya.
- **Finalists** at the **"Predicta v1.0" 2023,** organized by IEEE Student Branch of University of Peradeniya.
- **Finalists** at the **"IEEE Electronic Design Competition" 2023,** Organized by IEEE Sri Lanaka Section.
- Received **High Distinction Pass** at the **"Australian National Chemistry Quiz (Senior Division)" 2018**.

SELECTED PROJECTS

Computer Vision Action Recognition 2026 - Present

Focused on developing a new algorithm to group activity recognition in long videos.

Concepts: History aware understanding, Vision Language Models.

HeritageAI 2025 - Present

Under Nvidia Academic Grant Program

Focused on developing a unified restoration pipeline for degraded Sri Lankan historical images using implicit neural representations (INRs), with current work focused on INR based modules for denoising, super resolution, and colorization

Concepts: Implicit Neural Representations (INRs), Image Restoration.

Denoising Diffusion Models 2025 - Present

Conducted research on denoising diffusion models, focusing on improving generative fidelity and extending diffusion frameworks to structured audio signal generation.

Key studies:

- **Improving Diffusion Probabilistic Models (2025 - Present):** Incorporated trace energy regularization into the diffusion training objective to address predicted-noise energy underestimation in standard DDPMs. Achieved measurable improvements in generative fidelity across evaluation metrics.
- **Diffusion for Audio Signals (2025):** Investigated temporal structure representations for diffusion-based audio generation. Conditioned models on signal-processing-based temporal features to enhance waveform stability, temporal coherence, and perceptual realism relative to DiffWave.

Concepts: Denoising Diffusion Probabilistic Models (DDPM), Generative Modeling, Isotropy Constraints, Audio Signal Processing, Temporal Representation Learning, DiffWave.

Multispectral Imaging for Agricultural and Environmental Monitoring 2023 - 2025

Conducted multispectral sensing research for environmental and industrial analysis, developing machine learning models for soil characterization and algal bloom detection.

Key studies:

- **Soil Moisture Content Estimation (2023 - 2024):** Built regression models to estimate moisture levels from spectral features.
- **Soil Classification and Composition Prediction (2024 - 2025):** Developed machine learning models to classify soil texture and predict composition.
- **Microcystis Cell Count and Toxicity Estimation (2025):** Designed regression models to detect harmful algal blooms.

Concepts: Multispectral Sensing, Feature Extraction, Regression and Classification Models, Neural Networks.

PixelFold (UG) 2024

Developed Python-based image and hybrid video compression pipelines using DCT, quantization, and entropy coding. Implemented block-based motion estimation and predictive frame coding, with quantization adjustment for target bitrates and reconstruction quality.

Concepts: Discrete Cosine Transform (DCT), Huffman Coding, Run-Length Encoding, Predictive Coding, Rate-Distortion Analysis.

EXPERIENCE

Research and Development Engineer 2025 November - present

Farbe Technologies

Developing AI-driven signal and image processing solutions for industrial applications.

Volunteer Researcher 2023 June - present

Multidisciplinary AI Research Center, University of Peradeniya

Served as a volunteer researcher at MARC, contributing to the following theoretical and applied machine learning and deep learning projects across two research groups:

- Diffusion-based Generative AI
- Multispectral Imaging for agricultural and environmental monitoring

Teaching Assistent 2025 August - 2025 October

Department of Electrical and Electronic Engineering, University of Peradeniya

Provided guidance to students in EE358 power module and EE1010 electricity module.

Trainee Research and Development Engineer 2023 July – 2023 October

SLT Digital Lab, Sri Lanka Telecom

Contributed to the 'Deep Learning for RACOON AI' project, working on object detection and tracking using YOLOv8, deploying machine learning models with Seldon Core, managing databases with PostgreSQL, and utilizing ONNX for model interoperability.

SERVICES

Public Speaker/Volunteer - AI Workshops 2025

Participated in workshops to share knowledge on generative AI and Mulitispectral Imgaing for school students through the Multidisciplinary AI Research Center at the University of Peradeniya.

Executive committee member, IET on Campus, University of Peradeniya 2022 June - 2023 May

Assisted in organizing technical events and workshops to undergraduates.

Reporting team member, IEEE Robotics and Automation Society, University of Peradeniya 2022

Supported the coordination of technical events and workshops to undergraduates, while also contributing to editorial content and documentation.

Team member, OGV/H4TF CXP, AIESEC in Kandy 2022

Engaged in tasks related to tracking customer experience in AIESEC projects.

Executive committee member, Astronomical Society of Mahamaya Girls' College, Kandy 2018-2019

Organized astronomical nights and workshops

SKILLS

- **Programming Languages** - Python, C, Assembly
- **Libraries frameworks** - TensorFlow, PyTorch, Scikit- Learn, Keras, ONNX, YOLOv8, cv2, Scipy
- **Research** - Academic research, Project documentation, Technical writing
- **Other** – Latex, Git, SQL,PostgreSQL

REFERENCES

Prof. Roshan Godaliyadda
Ph.D.(National University of Singapore)

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